

## WAGGA WAGGA, Australia

WMO: 949100

Lat: 35.1583S

Lon: 147.4572E

Elev: 213

StdP: 98.79

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | -0.4       | 0.8 | -6.8                          | 2.2 | 26.5 | -4.1 | 2.7 | 22.5 | 10.1                  | 9.7  | 9.0 | 10.8 | 1.3                   | 110  | 0.487 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 14.5                   | 37.9            | 18.5 | 35.7 | 18.2 | 33.7 | 17.8 | 21.4                | 29.3 | 20.6 | 28.7 | 19.9 | 28.3 | 5.2                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 19.8                            | 14.9 | 23.0 | 18.4 | 13.7 | 22.5 | 17.2 | 12.6 | 22.4 | 63.3          | 29.2 | 60.4 | 28.5 | 57.8 | 28.4 | 25.3           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 9.9           | 8.7  | 7.7 | DB                         | -2.7 | 41.0               | 0.9 | 2.2                                                | -3.3 | 42.6       | -3.8 | 43.9       | -4.3 | 45.1       | -4.9 | 46.7 | (4) |
| (5)               |      |     | WB                         | -2.9 | 22.8               | 0.8 | 1.1                                                | -3.5 | 23.6       | -4.0 | 24.3       | -4.4 | 24.9       | -5.0 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 16.0   | 25.0  | 24.0  | 20.7  | 16.2  | 11.9  | 9.2   | 8.2   | 9.0   | 11.8  | 15.3  | 19.3  | 22.2  | (6)  |  |
|                                                                        | DBStd       | 6.73   | 4.03  | 3.76  | 3.61  | 3.36  | 2.96  | 2.44  | 2.22  | 2.43  | 2.95  | 3.77  | 4.18  | 4.05  | (7)  |  |
|                                                                        | HDD10.0     | 186    | 0     | 0     | 0     | 1     | 14    | 44    | 64    | 48    | 13    | 2     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 1536   | 2     | 3     | 18    | 79    | 202   | 275   | 314   | 288   | 198   | 109   | 37    | 10    | (9)  |  |
|                                                                        | CDD10.0     | 2383   | 464   | 392   | 332   | 187   | 72    | 19    | 8     | 18    | 66    | 167   | 279   | 378   | (10) |  |
|                                                                        | CDD18.3     | 691    | 208   | 162   | 91    | 16    | 1     | 0     | 0     | 0     | 1     | 16    | 66    | 130   | (11) |  |
|                                                                        | CDH23.3     | 8386   | 2623  | 1802  | 946   | 174   | 3     | 0     | 0     | 0     | 22    | 267   | 856   | 1693  | (12) |  |
|                                                                        | CDH26.7     | 4004   | 1435  | 883   | 366   | 35    | 0     | 0     | 0     | 0     | 4     | 84    | 371   | 826   | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.4    | 4.2   | 3.8   | 3.5   | 2.9   | 2.7   | 2.7   | 2.8   | 3.0   | 3.3   | 3.6   | 3.9   | 4.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 577    | 43    | 39    | 45    | 41    | 51    | 48    | 55    | 53    | 51    | 54    | 47    | 49    | (15) |  |
|                                                                        | PrecMax     | 1006   | 174   | 188   | 208   | 217   | 184   | 139   | 130   | 101   | 168   | 182   | 151   | 213   | (16) |  |
|                                                                        | PrecMin     | 244    | 0     | 0     | 0     | 0     | 4     | 1     | 2     | 6     | 5     | 3     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 164    | 34    | 39    | 46    | 38    | 41    | 29    | 29    | 26    | 33    | 41    | 35    | 40    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 41.2  | 39.8  | 35.7  | 30.1  | 23.7  | 19.1  | 17.2  | 20.7  | 26.0  | 32.1  | 37.0  | 39.1  | (19) |  |
|                                                                        |             | MCWB   | 19.5  | 19.5  | 18.2  | 16.3  | 14.3  | 12.3  | 11.8  | 13.4  | 13.8  | 15.2  | 17.3  | 17.6  | (20) |  |
|                                                                        | 2%          | DB     | 38.5  | 36.5  | 32.6  | 27.2  | 21.2  | 16.8  | 15.4  | 18.1  | 22.9  | 29.1  | 33.6  | 36.0  | (21) |  |
|                                                                        |             | MCWB   | 19.1  | 18.8  | 17.5  | 15.7  | 13.5  | 11.6  | 11.0  | 11.9  | 13.8  | 15.3  | 16.5  | 17.7  | (22) |  |
|                                                                        | 5%          | DB     | 36.1  | 34.0  | 30.4  | 25.2  | 19.7  | 15.4  | 14.1  | 16.2  | 20.8  | 26.2  | 30.9  | 33.5  | (23) |  |
|                                                                        |             | MCWB   | 18.6  | 18.4  | 17.2  | 15.0  | 13.0  | 11.2  | 10.1  | 10.9  | 13.1  | 14.6  | 16.3  | 17.1  | (24) |  |
|                                                                        | 10%         | DB     | 33.8  | 31.9  | 28.5  | 23.2  | 17.9  | 14.0  | 12.9  | 14.7  | 18.8  | 23.7  | 28.2  | 31.2  | (25) |  |
|                                                                        |             | MCWB   | 18.2  | 18.0  | 16.6  | 14.3  | 12.4  | 10.6  | 9.6   | 10.2  | 12.5  | 14.1  | 15.7  | 16.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 22.3  | 23.2  | 21.4  | 18.7  | 16.7  | 14.2  | 13.5  | 14.4  | 16.6  | 18.7  | 20.5  | 21.3  | (27) |  |
|                                                                        |             | MCDB   | 31.3  | 27.5  | 27.0  | 24.6  | 18.6  | 16.1  | 15.2  | 18.4  | 21.6  | 24.6  | 28.0  | 27.5  | (28) |  |
|                                                                        | 2%          | WB     | 21.4  | 21.6  | 20.0  | 17.2  | 15.1  | 12.9  | 11.8  | 12.9  | 15.4  | 17.3  | 19.3  | 20.1  | (29) |  |
|                                                                        |             | MCDB   | 30.4  | 28.8  | 25.9  | 22.7  | 18.9  | 15.3  | 14.4  | 16.7  | 20.2  | 23.4  | 26.5  | 27.3  | (30) |  |
|                                                                        | 5%          | WB     | 20.6  | 20.6  | 19.0  | 16.3  | 14.0  | 11.9  | 10.8  | 11.6  | 14.2  | 16.2  | 18.2  | 19.1  | (31) |  |
|                                                                        |             | MCDB   | 30.1  | 28.3  | 26.0  | 22.1  | 18.2  | 14.6  | 13.2  | 15.5  | 18.8  | 22.5  | 25.1  | 27.6  | (32) |  |
|                                                                        | 10%         | WB     | 19.8  | 19.7  | 18.1  | 15.3  | 13.0  | 11.0  | 9.9   | 10.5  | 13.0  | 15.2  | 17.2  | 18.2  | (33) |  |
|                                                                        |             | MCDB   | 30.0  | 27.8  | 25.4  | 21.7  | 17.0  | 13.5  | 12.2  | 14.1  | 17.6  | 21.6  | 25.0  | 27.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 14.5  | 13.5  | 13.5  | 13.1  | 11.4  | 9.5   | 9.3   | 10.8  | 12.6  | 14.1  | 14.2  | 14.9  | (35) |  |
|                                                                        | 5% DB       | MCDBR  | 17.0  | 15.8  | 15.6  | 15.1  | 13.2  | 11.6  | 11.2  | 13.4  | 15.6  | 17.8  | 17.7  | 17.2  | (36) |  |
|                                                                        |             | MCWBR  | 5.1   | 4.9   | 5.5   | 6.3   | 6.7   | 7.0   | 7.1   | 7.9   | 8.0   | 7.2   | 5.8   | 5.4   | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 13.6  | 11.9  | 11.8  | 11.9  | 10.5  | 9.2   | 9.1   | 11.7  | 12.2  | 13.2  | 13.0  | 13.2  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.359 | 0.344 | 0.323 | 0.309 | 0.299 | 0.289 | 0.285 | 0.295 | 0.320 | 0.327 | 0.345 | 0.343 | (40) |  |
|                                                                        | taud        |        | 2.433 | 2.512 | 2.550 | 2.552 | 2.542 | 2.586 | 2.597 | 2.541 | 2.459 | 2.463 | 2.455 | 2.476 | (41) |  |
|                                                                        | Ebn at Noon |        | 976   | 970   | 955   | 911   | 864   | 849   | 875   | 914   | 941   | 976   | 984   | 997   | (42) |  |
|                                                                        | Edh at Noon |        | 122   | 108   | 97    | 85    | 75    | 67    | 70    | 84    | 104   | 112   | 118   | 117   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.68  | 6.70  | 5.66  | 4.20  | 2.96  | 2.24  | 2.45  | 3.36  | 4.78  | 6.17  | 7.00  | 7.74  | (44) |  |
|                                                                        | RadStd      |        | 0.41  | 0.44  | 0.36  | 0.21  | 0.16  | 0.22  | 0.14  | 0.28  | 0.44  | 0.50  | 0.46  | 0.30  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |
| (46) | Station Only            | +0.49 | N/A     | N/A    | +1.28   | +0.39 | +0.51 | N/A         | N/A     | N/A     | N/A     | (46) |
| (47) | Regional (22 neighbors) | +0.40 | N/A     | N/A    | +1.05   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | (47) |

Nomenclature: See separate page